Practical 1

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Ex.1 Selected 2 Motors

Items	Value	Probability
Total Motors	10	
Good Motors	8	
Defective Motors	2	

*Find the probability	P()	value	Equation	Answer
[1] Both good motors P(A)	n(A)	28	P(A)=n(A)/n(S)	0.622222222
	n(S)	45		
[2] one motor is good and one motor is defective P(B)	n(B)	16	P(B)=n(B)/n(S)	0.355555556
	n(S)	45		

Ex.2 Rating probability

Rating	P()	value
Very Poor	P(A)	0.07
Poor	P(B)	0.12
Fair	P(C)	0.17
Good	P(D)	0.32
Very Good	P(E)	0.21
Excellent	P(F)	0.11

*Find the probability		Equation	Answer
[1] Very poor,Poor Fair or Good	P(G)	P(G)=P(A)+P(B)+P(C)+P(D)	0.68
[2] Good,Very Good, Excellent	P(H)	P(H)=P(D)+P(E)+P(F)	0.64

Ex.3 Warranty Repair

Event	P()	value
Engine Repair	P(A)	0.87
Driven train	P(B)	0.36
Both Pair	P(A andB)	0.29

*Find the probability	Equation	Answer
[1] one or both or another P(A or B)	**P(A or B) = P(A) +P(B) -P(A and B)**	0.94

Ex.4 Two Balance Dice

Total num of pair n(S) 36

*Find the probability	n()	value	Equation	Answer
[1] getting sum of 7 P(A)	n(A)	6	P(A)=n(A)/n(S)	0.166666667
[2] getting sum of 11 P(B)	n(B)	2	P(B)=n(B)/n(S)	0.055555556
[3] getting sum of 7or 11 P(C)	n(C)	8	P(C)=n(C)/n(S)	0.222222222
[4] getting sum of 3 P(D)	n(D)	2	P(D)=n(D)/n(S)	0.055555556
[5] getting sum of 2 or 12	n(E)	2	P(E)=n(E)/n(S)	0.055555556
[6] getting sum of 2,3 or 12	n(F)	4	P(F)=n(F)/n(S)	0.111111111

Ex.5 Incomplete Repair

Technician	P(Technician)	P(Incomplete/Technician)	Joint Probability
Janet	0.2	0.05	0.01
Tom	0.6	0.1	0.06
Georgia	0.15	0.1	0.015
Peter	0.05	0.05	0.0025

Total probability of joint Probability	0.0875

*Find the probability	Equation	Answer
P(Janet/ Incomplete repair)	P(Janet and Incomplete repair) / P(Incomplete)	0.114285714